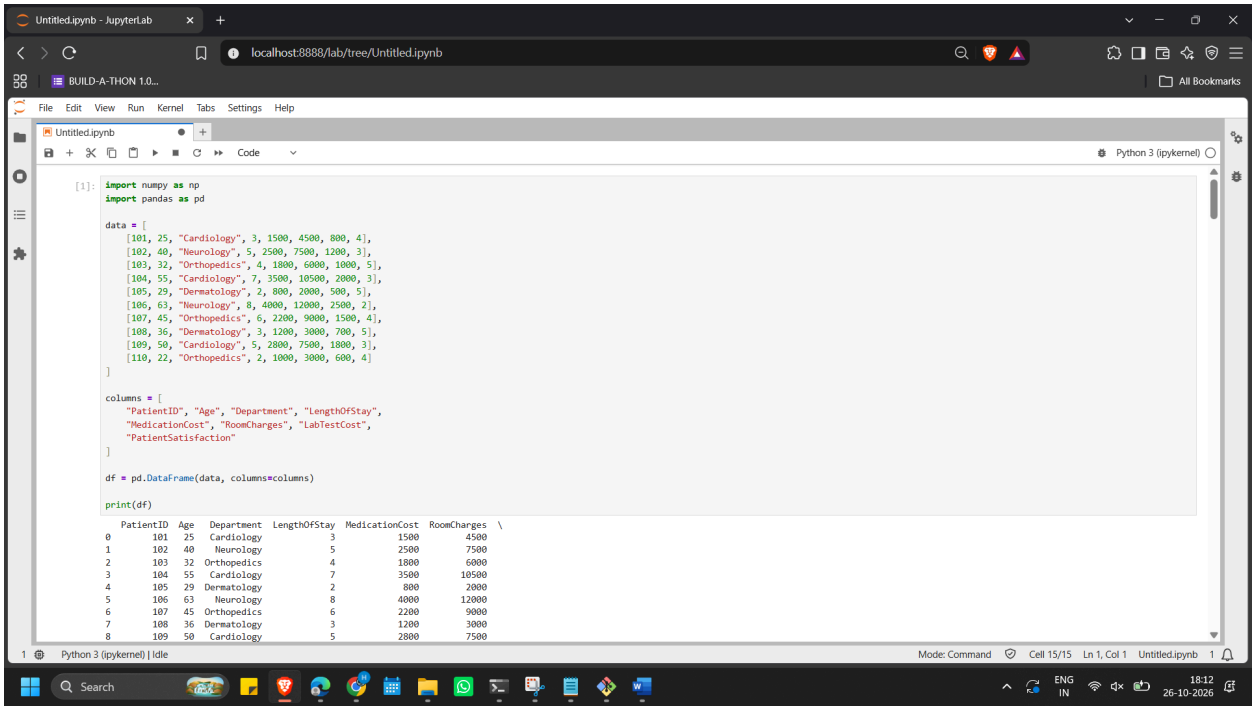
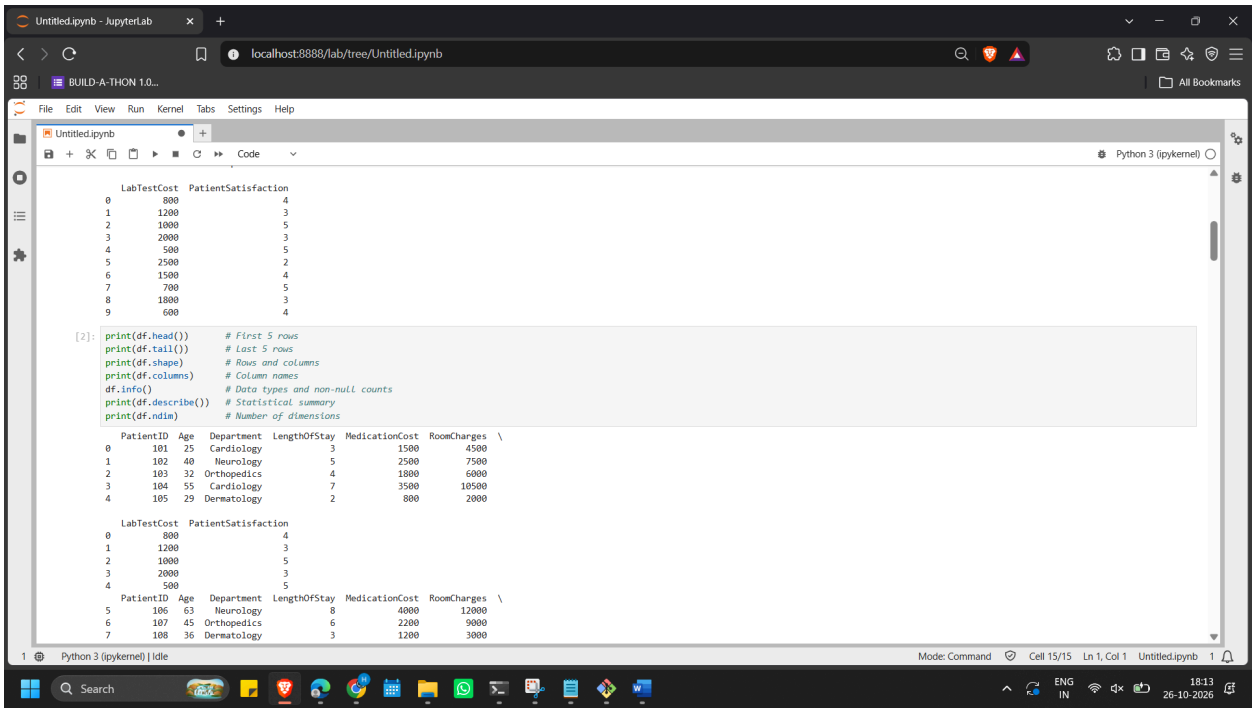
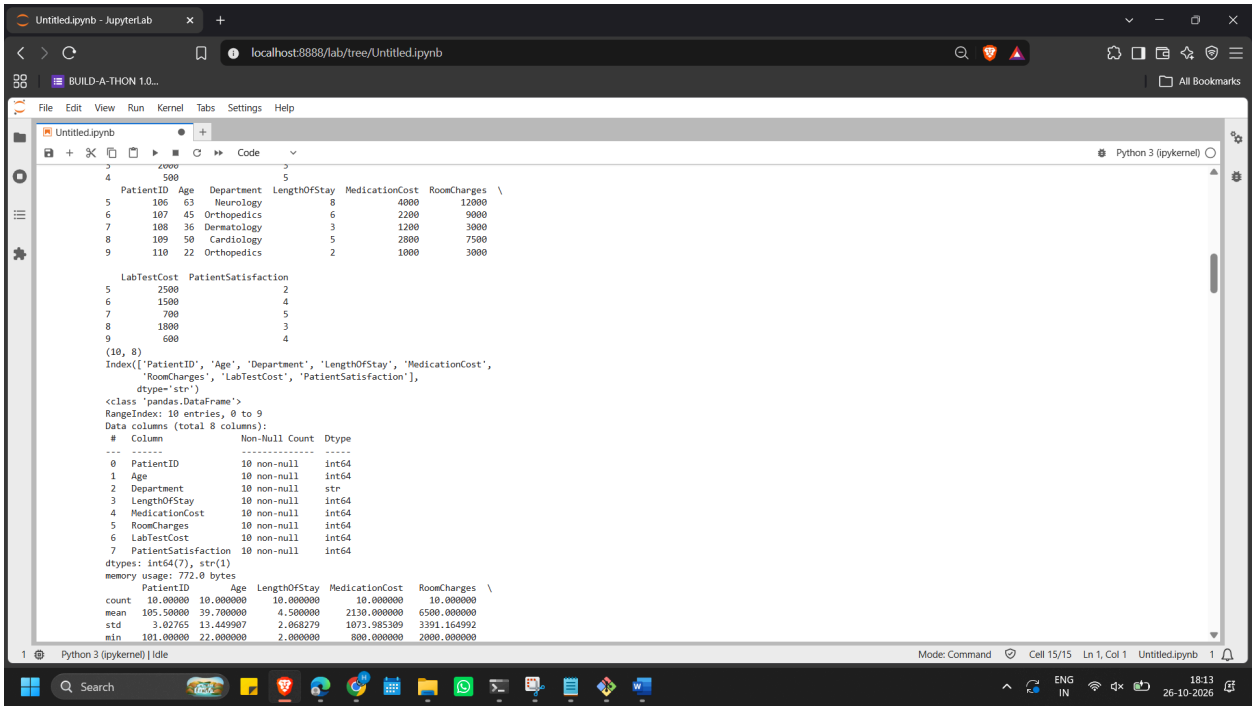

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Untitled.ipynb
Code
Python 3 (ipykernel)
[1]:
import numpy as np
import pandas as pd
data = [
[101, 25, "Cardiology", 3, 1500, 4500, 800, 4],
[102, 40, "Neurology", 5, 2500, 7500, 1200, 3],
[103, 32, "Orthopedics", 4, 1800, 6000, 1000, 5],
[104, 55, "Cardiology", 7, 3500, 10500, 2000, 3],
[105, 29, "Dermatology", 2, 800, 2000, 500, 5],
[106, 63, "Neurology", 8, 4000, 12000, 2500, 2],
[107, 45, "Orthopedics", 6, 2200, 9000, 1500, 4],
[108, 36, "Dermatology", 3, 1200, 3000, 700, 5],
[109, 50, "Cardiology", 5, 2800, 7500, 1800, 3],
[110, 22, "Orthopedics", 2, 1000, 3000, 600, 4]
]
columns = [
"PatientID", "Age", "Department", "LengthOfStay",
"MedicationCost", "RoomCharges", "LabTestCost",
"PatientSatisfaction"
]
df = pd.DataFrame(data, columns=columns)
print(df)
PatientID Age Department LengthOfStay MedicationCost RoomCharges \
0 101 25 Cardiology 3 1500 4500
1 102 40 Neurology 5 2500 7500
2 103 32 Orthopedics 4 1800 6000
3 104 55 Cardiology 7 3500 10500
4 105 29 Dermatology 2 800 2000
5 106 63 Neurology 8 4000 12000
6 107 45 Orthopedics 6 2200 9000
7 108 36 Dermatology 3 1200 3000
8 109 50 Cardiology 5 2800 7500
Python 3 (ipykernel) | Idle
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Untitled.ipynb
Code
Python 3 (ipykernel)
LabTestCost PatientSatisfaction
0 800 4
1 1200 3
2 1000 5
3 2000 3
4 500 5
5 2500 2
6 1500 4
7 700 5
8 1800 3
9 600 4
[2]:
print(df.head()) # First 5 rows
print(df.tail()) # Last 5 rows
print(df.shape) # Rows and columns
print(df.columns) # Column names
df.info() # Data types and non-null counts
print(df.describe()) # Statistical summary
print(df.ndim) # Number of dimensions
PatientID Age Department LengthOfStay MedicationCost RoomCharges \
0 101 25 Cardiology 3 1500 4500
1 102 40 Neurology 5 2500 7500
2 103 32 Orthopedics 4 1800 6000
3 104 55 Cardiology 7 3500 10500
4 105 29 Dermatology 2 800 2000
LabTestCost PatientSatisfaction
0 800 4
1 1200 3
2 1000 5
3 2000 3
4 500 5
PatientID Age Department LengthOfStay MedicationCost RoomCharges \
5 106 63 Neurology 8 4000 12000
6 107 45 Orthopedics 6 2200 9000
7 108 36 Dermatology 3 1200 3000
Python 3 (ipykernel) | Idle
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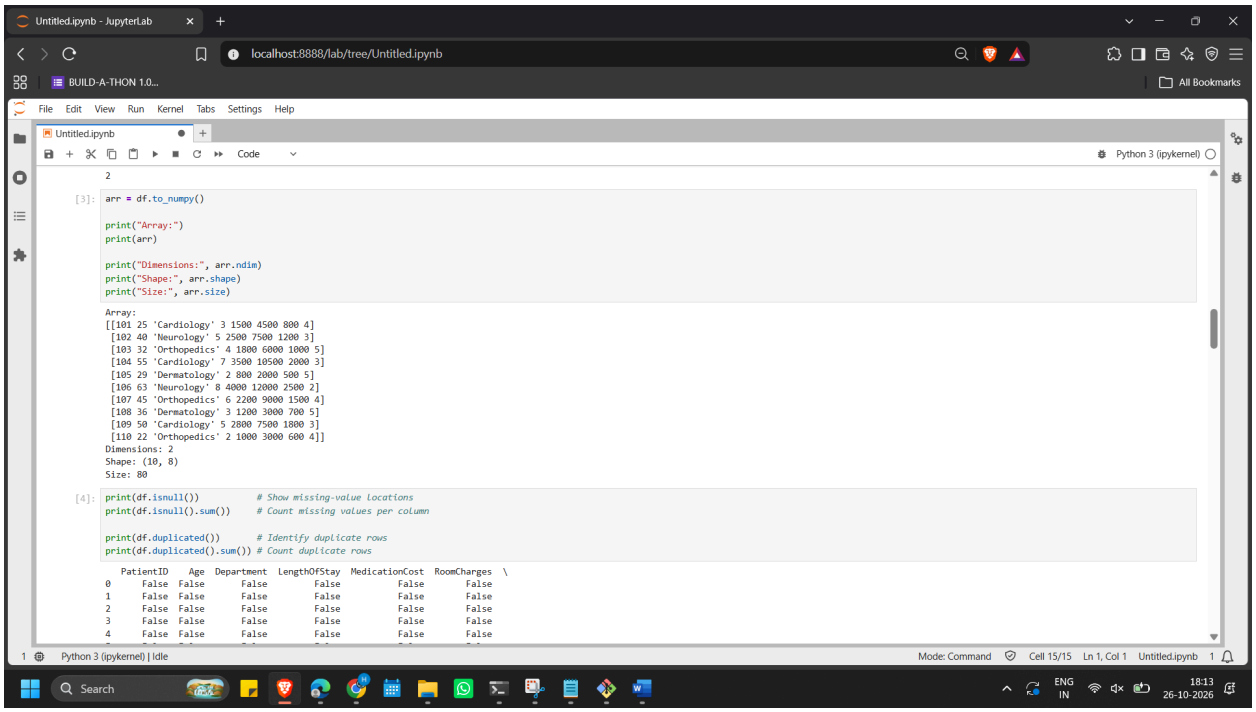


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Untitled.ipynb
Code
Python 3 (ipykernel)

4         500                5
   PatientID  Age   Department  LengthOfStay  MedicationCost  RoomCharges  \
5        106   63    Neurology             8            4000        12000
6        107   45  Orthopedics             6            2200         9000
7        108   36  Dermatology             3            1200         3000
8        109   50   Cardiology             5            2800         7500
9        110   22  Orthopedics             2            1000         3000

   LabTestCost  PatientSatisfaction
5         2500                    2
6         1500                    4
7          700                    5
8         1800                    3
9          600                    4
(10, 8)
Index(['PatientID', 'Age', 'Department', 'LengthOfStay', 'MedicationCost',
       'RoomCharges', 'LabTestCost', 'PatientSatisfaction'],
      dtype='str')
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 8 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   PatientID            10 non-null     int64
 1   Age                  10 non-null     int64
 2   Department           10 non-null     str
 3   LengthOfStay         10 non-null     int64
 4   MedicationCost       10 non-null     int64
 5   RoomCharges          10 non-null     int64
 6   LabTestCost          10 non-null     int64
 7   PatientSatisfaction  10 non-null     int64
dtypes: int64(7), str(1)
memory usage: 772.0 bytes
       PatientID        Age  LengthOfStay  MedicationCost   RoomCharges  \
count   10.00000  10.000000     10.000000       10.000000    10.000000
mean   105.50000  39.700000      4.500000     2130.000000  6500.000000
std      3.02765  13.449907      2.068279     1073.985309  3391.164992
min    101.00000  22.000000      2.000000      800.000000  2000.000000

Python 3 (ipykernel) | Idle
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Untitled.ipynb - JupyterLab
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Untitled.ipynb
Code
Python 3 (ipykernel)

2

[3]:
arr = df.to_numpy()

print("Array:")
print(arr)

print("Dimensions:", arr.ndim)
print("Shape:", arr.shape)
print("Size:", arr.size)

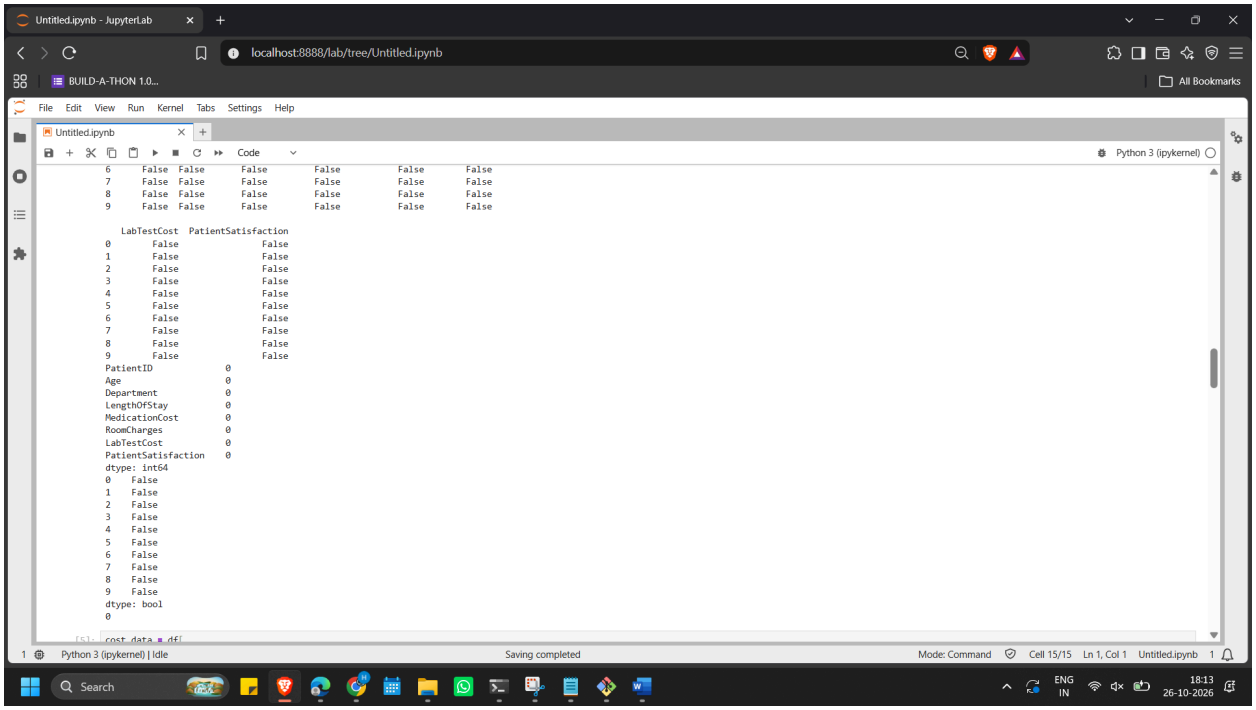
Array:
[[101 25 'Cardiology' 3 1500 4500 800 4]
 [102 40 'Neurology' 5 2500 7500 1200 3]
 [103 32 'Orthopedics' 4 1800 6000 1000 5]
 [104 55 'Cardiology' 7 3500 10500 2000 3]
 [105 29 'Dermatology' 2 800 2000 500 5]
 [106 63 'Neurology' 8 4000 12000 2500 2]
 [107 45 'Orthopedics' 6 2200 9000 1500 4]
 [108 36 'Dermatology' 3 1200 3000 700 5]
 [109 50 'Cardiology' 5 2800 7500 1800 3]
 [110 22 'Orthopedics' 2 1000 3000 600 4]]
Dimensions: 2
Shape: (10, 8)
Size: 80

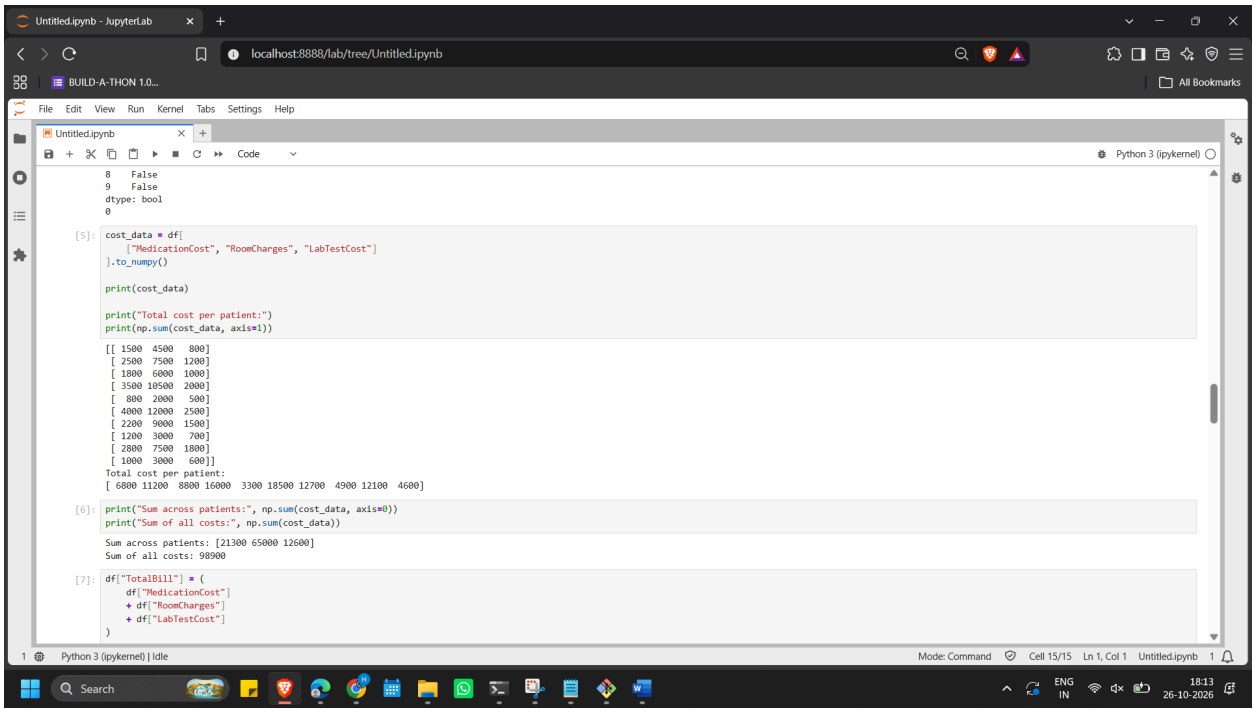
[4]:
print(df.isnull())          # Show missing-value locations
print(df.isnull().sum())    # Count missing values per column

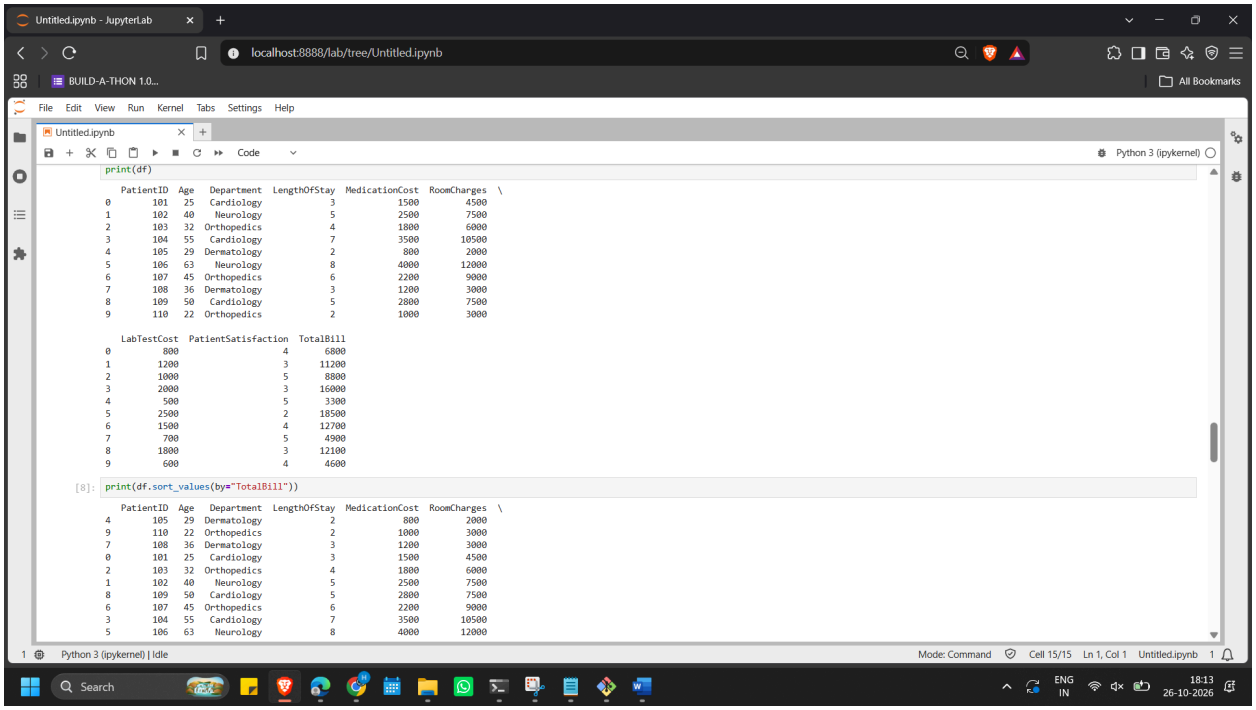
print(df.duplicated())      # Identify duplicate rows
print(df.duplicated().sum()) # Count duplicate rows

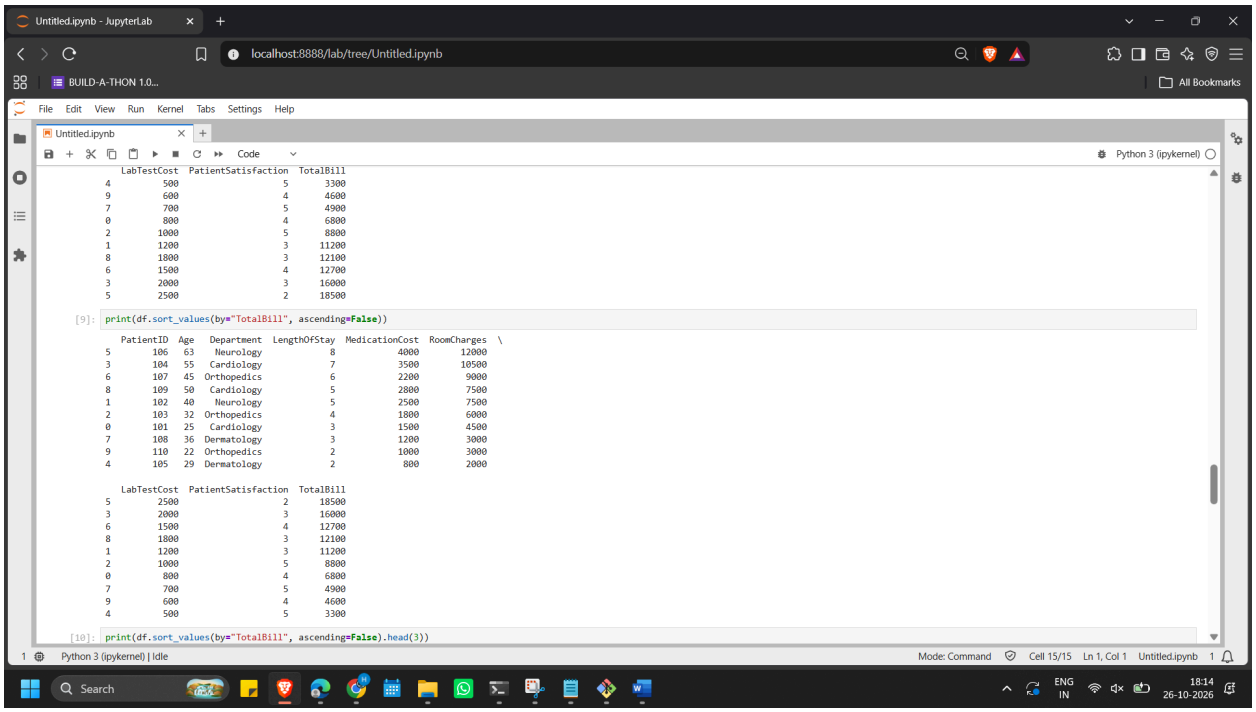
   PatientID    Age  Department  LengthOfStay  MedicationCost  RoomCharges  \
0      False  False       False         False           False        False
1      False  False       False         False           False        False
2      False  False       False         False           False        False
3      False  False       False         False           False        False
4      False  False       False         False           False        False

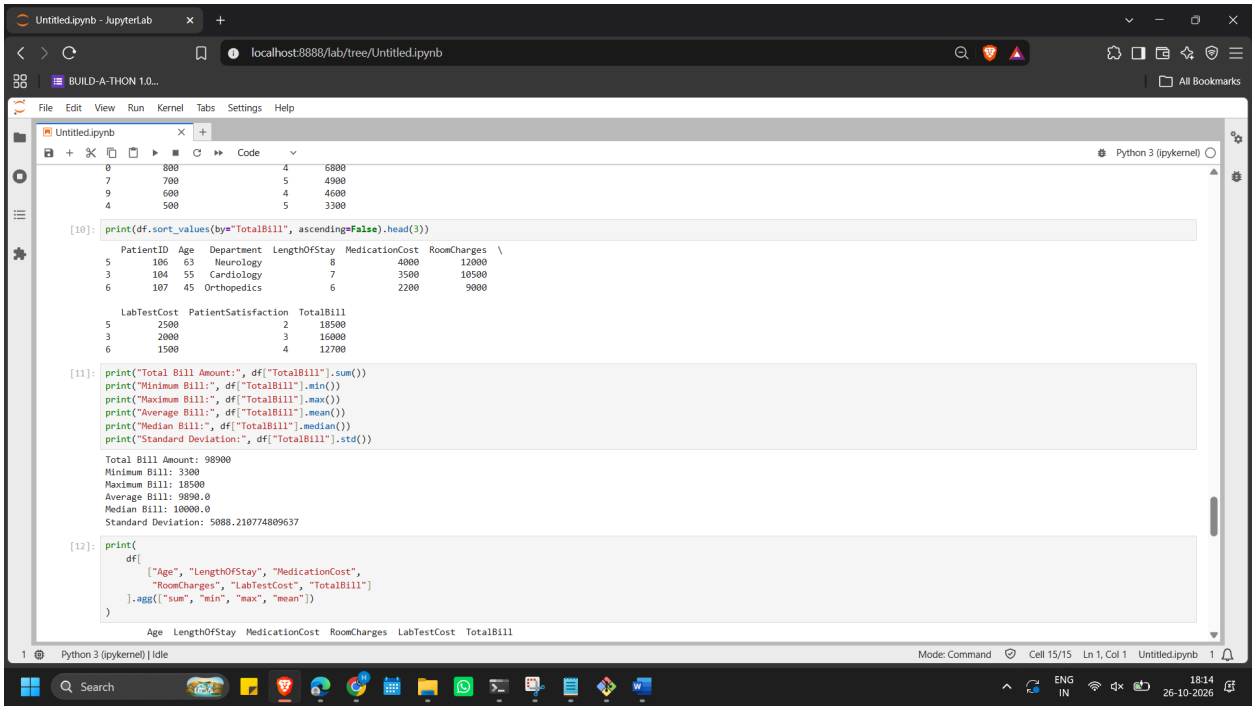
Python 3 (ipykernel) | Idle
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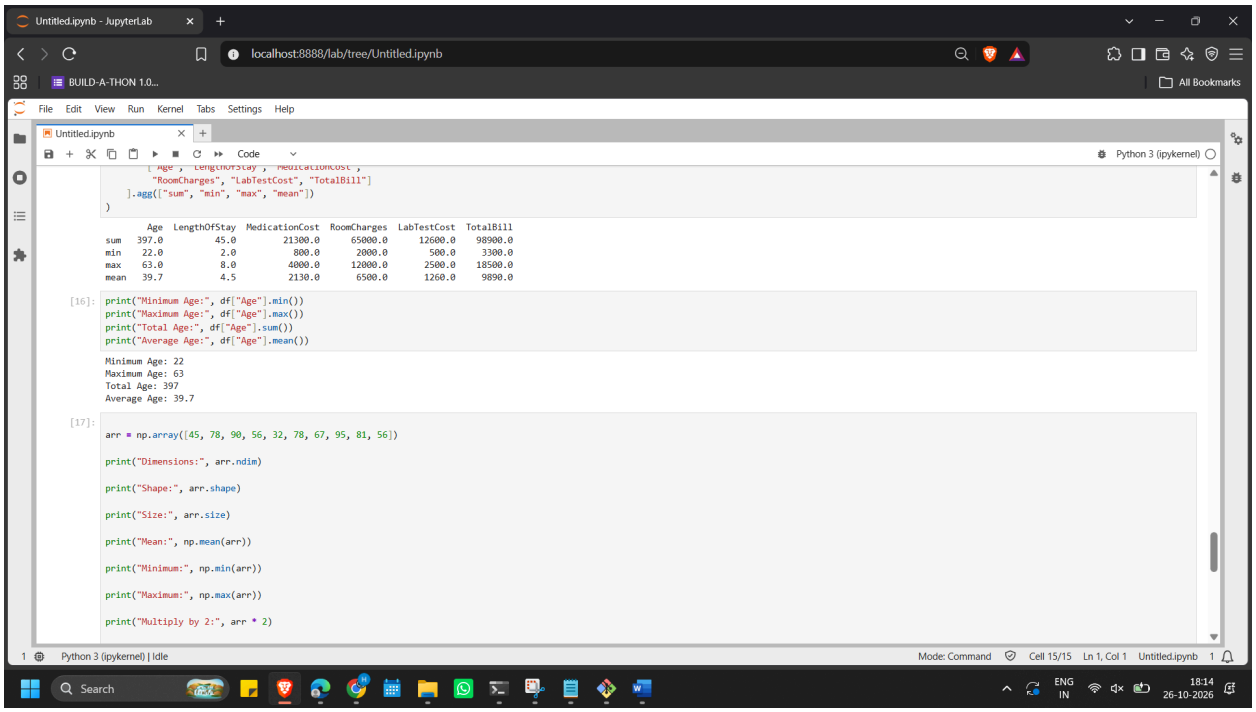

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File Edit View Run Kernel Tabs Settings Help
Untitled.ipynb
Code
Python 3 (ipykernel)
6 False False False False False False
7 False False False False False False
8 False False False False False False
9 False False False False False False
LabTestCost PatientSatisfaction
0 False False
1 False False
2 False False
3 False False
4 False False
5 False False
6 False False
7 False False
8 False False
9 False False
PatientID 0
Age 0
Department 0
LengthOfStay 0
MedicationCost 0
RoomCharges 0
LabTestCost 0
PatientSatisfaction 0
dtype: int64
0 False
1 False
2 False
3 False
4 False
5 False
6 False
7 False
8 False
9 False
dtype: bool
0
[5]: cost_data = df[
Python 3 (ipykernel) | Idle
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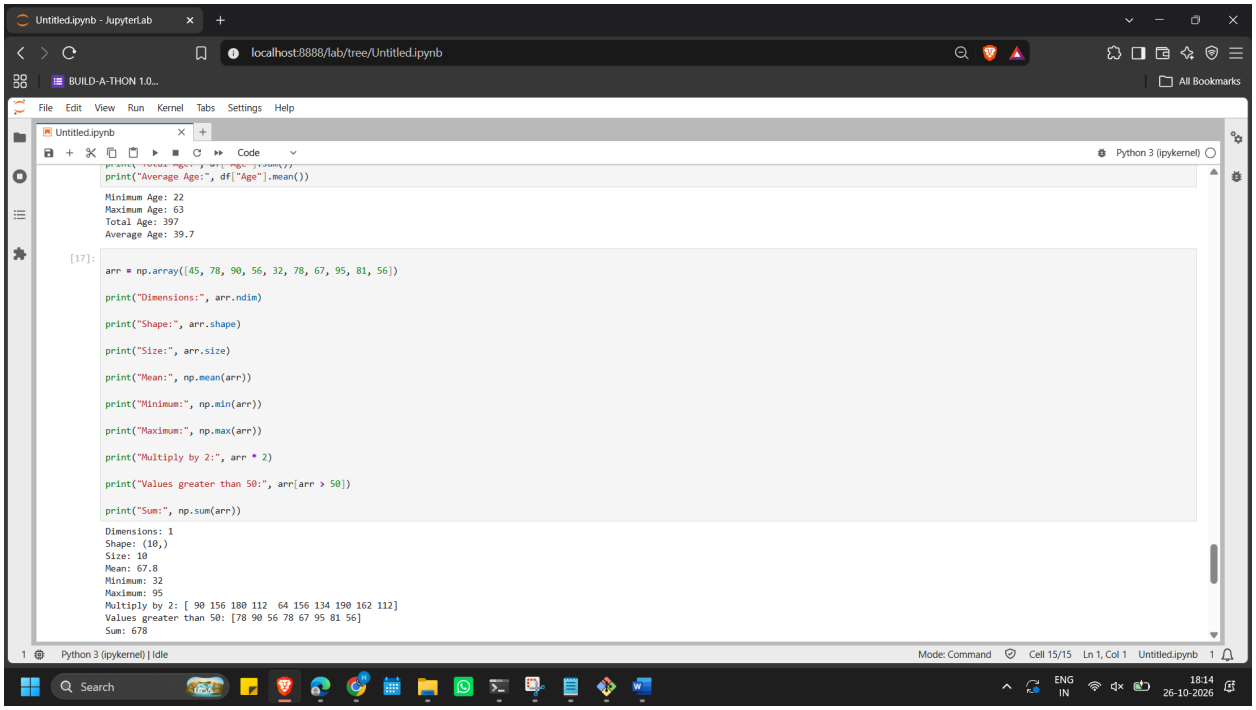

Untitled.ipynb - JupyterLab
localhost:8888/lab/tree/Untitled.ipynb
File Edit View Run Kernel Tabs Settings Help
Untitled.ipynb
Code
Python 3 (ipykernel)
8 False
9 False
dtype: bool
0
[5]: cost_data = df[
["MedicationCost", "RoomCharges", "LabTestCost"]
].to_numpy()
print(cost_data)
print("Total cost per patient:")
print(np.sum(cost_data, axis=1))
[[1500 4500 800]
[2500 7500 1200]
[1800 6000 1000]
[3500 10500 2000]
[800 2000 500]
[4000 12000 2500]
[2200 9000 1500]
[1200 3000 700]
[2800 7500 1800]
[1000 3000 600]]
Total cost per patient:
[6800 11200 8800 16000 3300 18500 12700 4900 12100 4600]
[6]: print("Sum across patients:", np.sum(cost_data, axis=0))
print("Sum of all costs:", np.sum(cost_data))
Sum across patients: [21300 65000 12600]
Sum of all costs: 98900
[7]: df["TotalBill"] = (
df["MedicationCost"]
+ df["RoomCharges"]
+ df["LabTestCost"]
)
Python 3 (ipykernel) | Idle
Mode: Command
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Untitled.ipynb - JupyterLab
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File Edit View Run Kernel Tabs Settings Help
Untitled.ipynb
Code
Python 3 (ipykernel)
print(df)
PatientID Age Department LengthOfStay MedicationCost RoomCharges \
0 101 25 Cardiology 3 1500 4500
1 102 40 Neurology 5 2500 7500
2 103 32 Orthopedics 4 1800 6000
3 104 55 Cardiology 7 3500 10500
4 105 29 Dermatology 2 800 2000
5 106 63 Neurology 8 4000 12000
6 107 45 Orthopedics 6 2200 9000
7 108 36 Dermatology 3 1200 3000
8 109 50 Cardiology 5 2800 7500
9 110 22 Orthopedics 2 1000 3000
LabTestCost PatientSatisfaction TotalBill
0 800 4 6800
1 1200 3 11200
2 1000 5 8800
3 2000 3 16000
4 500 5 3300
5 2500 2 18500
6 1500 4 12700
7 700 5 4900
8 1800 3 12100
9 600 4 4600
[8]: print(df.sort_values(by="TotalBill"))
PatientID Age Department LengthOfStay MedicationCost RoomCharges \
4 105 29 Dermatology 2 800 2000
9 110 22 Orthopedics 2 1000 3000
7 108 36 Dermatology 3 1200 3000
0 101 25 Cardiology 3 1500 4500
2 103 32 Orthopedics 4 1800 6000
1 102 40 Neurology 5 2500 7500
8 109 50 Cardiology 5 2800 7500
6 107 45 Orthopedics 6 2200 9000
3 104 55 Cardiology 7 3500 10500
5 106 63 Neurology 8 4000 12000
Python 3 (ipykernel) | Idle
Mode: Command
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File Edit View Run Kernel Tabs Settings Help
Untitled.ipynb
Code
Python 3 (ipykernel)
LabTestCost PatientSatisfaction TotalBill
4 500 5 3300
9 600 4 4600
7 700 5 4900
0 800 4 6800
2 1000 5 8800
1 1200 3 11200
8 1800 3 12100
6 1500 4 12700
3 2000 3 16000
5 2500 2 18500
[9]: print(df.sort_values(by="TotalBill", ascending=False))
PatientID Age Department LengthOfStay MedicationCost RoomCharges \
5 106 63 Neurology 8 4000 12000
3 104 55 Cardiology 7 3500 10500
6 107 45 Orthopedics 6 2200 9000
8 109 50 Cardiology 5 2800 7500
1 102 40 Neurology 5 2500 7500
2 103 32 Orthopedics 4 1800 6000
0 101 25 Cardiology 3 1500 4500
7 108 36 Dermatology 3 1200 3000
9 110 22 Orthopedics 2 1000 3000
4 105 29 Dermatology 2 800 2000
LabTestCost PatientSatisfaction TotalBill
5 2500 2 18500
3 2000 3 16000
6 1500 4 12700
8 1800 3 12100
1 1200 3 11200
2 1000 5 8800
0 800 4 6800
7 700 5 4900
9 600 4 4600
4 500 5 3300
[10]: print(df.sort_values(by="TotalBill", ascending=False).head(3))
Python 3 (ipykernel) | Idle
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File Edit View Run Kernel Tabs Settings Help
Untitled.ipynb
Code
Python 3 (ipykernel)
0 800 4 6800
7 700 5 4900
9 600 4 4600
4 500 5 3300
[10]: print(df.sort_values(by="TotalBill", ascending=False).head(3))
PatientID Age Department LengthOfStay MedicationCost RoomCharges \
5 106 63 Neurology 8 4000 12000
3 104 55 Cardiology 7 3500 10500
6 107 45 Orthopedics 6 2200 9000
LabTestCost PatientSatisfaction TotalBill
5 2500 2 18500
3 2000 3 16000
6 1500 4 12700
[11]: print("Total Bill Amount:", df["TotalBill"].sum())
print("Minimum Bill:", df["TotalBill"].min())
print("Maximum Bill:", df["TotalBill"].max())
print("Average Bill:", df["TotalBill"].mean())
print("Median Bill:", df["TotalBill"].median())
print("Standard Deviation:", df["TotalBill"].std())
Total Bill Amount: 98900
Minimum Bill: 3300
Maximum Bill: 18500
Average Bill: 9890.0
Median Bill: 10000.0
Standard Deviation: 5088.210774809637
[12]: print(
df[
["Age", "LengthOfStay", "MedicationCost",
"RoomCharges", "LabTestCost", "TotalBill"]
].agg(["sum", "min", "max", "mean"])
)
Age LengthOfStay MedicationCost RoomCharges LabTestCost TotalBill
1 Python 3 (ipykernel) | Idle
Mode: Command
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File Edit View Run Kernel Tabs Settings Help
Untitled.ipynb
Code
Python 3 (ipykernel)
"RoomCharges", "LabTestCost", "TotalBill"]
].agg(["sum", "min", "max", "mean"])
)
Age LengthOfStay MedicationCost RoomCharges LabTestCost TotalBill
sum 397.0 45.0 21300.0 65000.0 12600.0 98900.0
min 22.0 2.0 800.0 2000.0 500.0 3300.0
max 63.0 8.0 4000.0 12000.0 2500.0 18500.0
mean 39.7 4.5 2130.0 6500.0 1260.0 9890.0
[16]: print("Minimum Age:", df["Age"].min())
print("Maximum Age:", df["Age"].max())
print("Total Age:", df["Age"].sum())
print("Average Age:", df["Age"].mean())
Minimum Age: 22
Maximum Age: 63
Total Age: 397
Average Age: 39.7
[17]: arr = np.array([45, 78, 90, 56, 32, 78, 67, 95, 81, 56])
print("Dimensions:", arr.ndim)
print("Shape:", arr.shape)
print("Size:", arr.size)
print("Mean:", np.mean(arr))
print("Minimum:", np.min(arr))
print("Maximum:", np.max(arr))
print("Multiply by 2:", arr * 2)
1 Python 3 (ipykernel) | Idle
Mode: Command
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File Edit View Run Kernel Tabs Settings Help
Untitled.ipynb
Code
Python 3 (ipykernel)
print("Average Age:", df["Age"].mean())
Minimum Age: 22
Maximum Age: 63
Total Age: 397
Average Age: 39.7
[17]:
arr = np.array([45, 78, 90, 56, 32, 78, 67, 95, 81, 56])
print("Dimensions:", arr.ndim)
print("Shape:", arr.shape)
print("Size:", arr.size)
print("Mean:", np.mean(arr))
print("Minimum:", np.min(arr))
print("Maximum:", np.max(arr))
print("Multiply by 2:", arr * 2)
print("Values greater than 50:", arr[arr > 50])
print("Sum:", np.sum(arr))
Dimensions: 1
Shape: (10,)
Size: 10
Mean: 67.8
Minimum: 32
Maximum: 95
Multiply by 2: [90 156 180 112 64 156 134 190 162 112]
Values greater than 50: [78 90 56 78 67 95 81 56]
Sum: 678
Python 3 (ipykernel) | Idle
Mode: Command
Cell 15/15
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